

Rodrigo Sierra

Course: Artificial Intel Resource Ref - ITAI 2277

Instructor: Professor Sitaram Ayyagari

Date: 04/29/2026

The Solution

A fully automated recruitment pipeline that handles candidate screening from email to shortlist, no manual intervention required until the final interview stage.

Resume Screening

Automatically extracts and validates resume data from incoming emails

AI Candidate Analysis

Uses LLMs to parse unstructured resumes into structured, comparable data

Intelligent Job Matching

Matches candidate profiles to open roles based on skills and experience

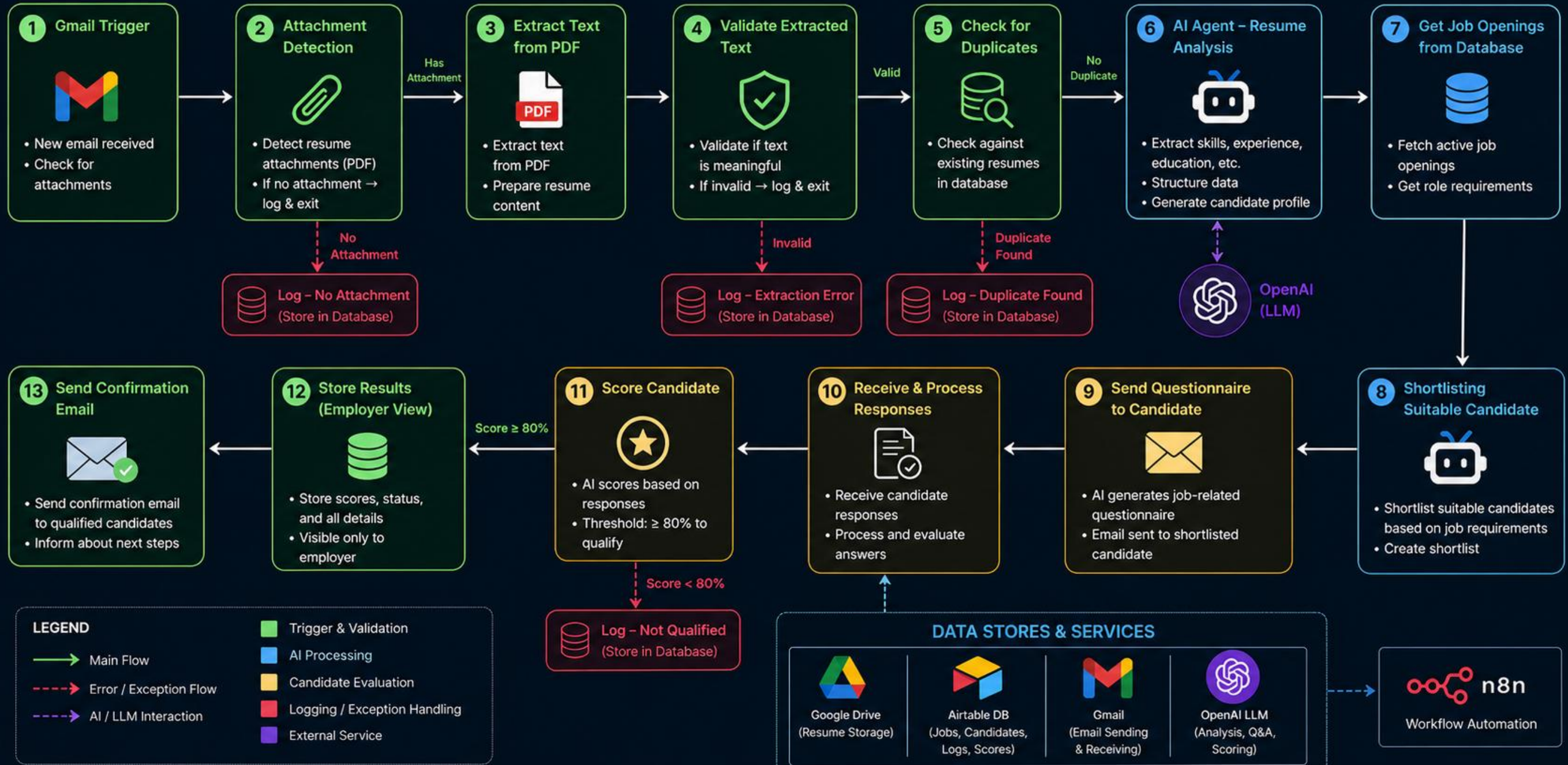
Smart Shortlisting

Candidates scoring $\geq 80\%$ on questionnaire are advanced to in-person interviews



AI-Powered Resume Screening & Job Matching System

Workflow Overview



Tools & Technologies



n8n

Orchestrates the entire workflow — connecting Gmail, Drive, Airtable, and OpenAI in a visual, no-code pipeline.



Gmail API

Triggers the pipeline on new emails and handles attachment extraction automatically.



Google Drive

Centralized, secure storage for all incoming resumes with seamless workflow integration.



Airtable

Stores candidate logs, job data, and duplicate detection records for full traceability.



OpenAI GPT-4o-mini (LLM)

Analyzes resumes, generates questionnaires, and scores candidate responses intelligently.

My AI Expertise



This project integrates several areas of artificial intelligence and applied technology. Natural Language Processing is used when the system reads resumes, extracts skills and experience, and evaluates candidate responses. Deep Learning is represented through the use of large language models, which understand and generate human language. Machine Learning concepts are applied through scoring, ranking, threshold-based shortlisting, and candidate-job matching. The project also demonstrates system integration by connecting Gmail, Google Drive, Airtable, OpenAI models, and n8n into one complete workflow.

What I learned



Overall, the project demonstrates the potential of AI in transforming recruitment processes.

This project also helped our team understand that solving real-world AI problems is not only about writing code or training a model from scratch. In many real situations, the value comes from knowing how to connect AI models, databases, APIs, and automation tools into a useful system. Through this project, we learned how to use n8n as an orchestrator, how to apply large language models to unstructured data, and how to design a workflow that supports recruiters while keeping human decision-making in the final step.

Thank You!